

Certified structural restrictions on $(5, 5)$ - and $(4, 6)$ -free graphs: an independently verifiable computational package

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Abstract

This note is a package of *independently verifiable computational artefacts* about both open families $43 \leq R(5, 5) \leq 46$ and $36 \leq R(4, 6) \leq 40$. Every statement below is backed by machine-checkable evidence: an exhaustive enumeration re-derived by a code-independent method, a DRAT unsatisfiability proof validated with the independent checker `drat-trim`, or a brute-force verifier whose logic is the defining condition itself. We report nothing as a search “finding”; searches without a certificate are explicitly delimited as such.

Concretely: (i) an exhaustive enumeration shows there is no *cyclic* (circulant) $(5, 5)$ -free colouring of K_n for $42 \leq n \leq 48$, while 20 of the 2^{20} cyclic colourings of K_{41} are $(5, 5)$ -free (explicit difference classes given); since a vertex-transitive graph of prime order is circulant, there is consequently *no vertex-transitive $(5, 5)$ -free graph on 43 or 47 vertices*. (ii) The same exhaustive treatment of cyclic families shows there is no cyclic $(4, 6)$ -free colouring of K_n for $34 \leq n \leq 40$, hence no vertex-transitive $(4, 6)$ -free graph on 37 vertices. (iii) The 42-vertex $(5, 5)$ -free graph of Exoo (1989) does not extend to 43 vertices: the associated 42-variable, 2318-clause unsatisfiability instance carries DRAT proofs validated externally. (iv) The two published witnesses (Exoo’s 42-vertex $(5, 5)$ -free graph and 35-vertex $(4, 6)$ -free graph) are re-verified from scratch by brute force as part of this package.

The negative statements are established by DRAT certificates; the positive colourings by an independent brute-force checker; all artefacts are listed with SHA-256 digests and a one-command verification protocol. We do not decide either $R(5, 5)$ or $R(4, 6)$. A 384-CPU-hour CDCL portfolio on K_{43} returned no verdict, and our simulated-annealing search — whose clique objective provably plateaus at cost > 0 regardless of parameters — failed to reproduce published constructions in equal budgets; both are reported as calibration results carrying no evidential weight.

1 Introduction

A two-colouring of the edges of a complete graph K_n is called (r, s) -free if it contains neither a red K_r nor a blue K_s ; equivalently $R(r, s) > n$. The diagonal case $r = s = 5$ has been open for more than three decades: writing $43 \leq R(5, 5) \leq 46$, the lower bound is Exoo’s 42-vertex construction of 1989 and the upper bound the counting-plus- machine-verification argument of Angelveit and McKay. The off-diagonal case $36 \leq R(4, 6) \leq 40$ is likewise open, with the lower bound due to Exoo (2012). The conjecture $R(5, 5) = 43$ is standard in the literature (Section 6).

Deciding the remaining gaps is out of reach of a single workstation: a rectangular K_{43} instance has 903 variables and $2 \cdot \binom{43}{5} = 1\,925\,196$ forbidden-clique clauses, and the searches reported below

spent hundreds of CPU-hours without a verdict. This note therefore does not attempt a decision. Its contribution is of a different kind: a *certified package of structural statements about highly symmetric families* — complete, cheap to establish, and verifiable by a reader with no access to our machines — which narrows the space in which any witness can live, together with an honest inventory of the non-certified searches.

The methodological stance is stated once, here. A computational claim is quoted in this note *only if* it is reproducible from the artefacts (Section 5) by an independent checker; otherwise it is labelled a non-result. In particular the negative statements are machine-checkable proofs, and the positive colourings are re-checked by brute force whose code is the definition itself.

Contribution 1 (Theorem 3). For every $n \in \{42, \dots, 48\}$ there is no cyclic $(5, 5)$ -free colouring of K_n ; for $n = 41$ such colourings exist and we exhibit them. Consequently (Corollary 4) there is no vertex-transitive $(5, 5)$ -free graph on 43 or 47 vertices.

Contribution 2 (Theorem 6). For every $n \in \{34, \dots, 40\}$ there is no cyclic $(4, 6)$ -free colouring of K_n . Consequently (Corollary 7) there is no vertex-transitive $(4, 6)$ -free graph on 37 vertices. The published 35-vertex $(4, 6)$ -free graph of Exoo is *not* cyclic.

Contribution 3 (Theorem 9). Exoo’s 42-vertex $(5, 5)$ -free graph admits no one-vertex extension to 43 vertices: it is maximal. The instance is certified unsatisfiable by DRAT proofs validated with `drat-trim`.

Contribution 4 (Proposition 11). Both published witnesses — Exoo’s 42-vertex $(5, 5)$ -free and 35-vertex $(4, 6)$ -free graphs — are re-verified from scratch by brute force as part of this package.

Section 2 fixes notation, Section 3 describes the methods and their validation, Section 4 states the results, Section 5 documents the complete set of reproducible artefacts, Section 6 positions the work relative to the literature, and Section 7 reports what did not work and why it carries no weight.

2 Definitions and notation

Definition 1. $R(r, s)$ is the least N such that every 2-colouring of $E(K_N)$ contains a red K_r or a blue K_s . A colouring with neither is called (r, s) -free. We identify a colouring with its set of red edges.

Definition 2. For $n \geq 3$, let $h = \lfloor n/2 \rfloor$. Label the vertices by $\mathbb{Z}_n$ and define the *chord class* of an unordered pair $\{i, j\}$ as $d = \min(|i - j| \bmod n, n - |i - j| \bmod n) \in \{1, \dots, h\}$. A colouring is *cyclic* (equivalently: a circulant colouring, or a colouring constant on the difference sets of $\mathbb{Z}_n$) if the colour of $\{i, j\}$ depends only on its chord class. A cyclic colouring is thus a subset $A \subseteq \{1, \dots, h\}$ of *red classes*, and there are exactly 2^h cyclic colourings of K_n .

We use the standard fact that a vertex-transitive graph of prime order is a circulant graph; equivalently, a transitive permutation group of prime degree is contained in the affine group, so a vertex-transitive graph on p vertices is determined by a set of differences in $\mathbb{Z}_p$.

3 Methods and their validation

Every statement in this note belongs to exactly one of two proof thesauri: (i) a positive statement rests on the brute-force verifier of Section 3.1; (ii) a negative statement rests on a dry run of an

exhaustive enumeration that is re-derived by a code-independent method and, for the families, certified by a DRAT proof (Sections 3.2–3.4). There is no third category: in particular no local-search output is ever quoted as evidence (Section 7).

3.1 Independent verifier for positive results

Every colouring reported as positive is checked by brute force: all $\binom{n}{r} + \binom{n}{s}$ subsets are enumerated and tested for being monochromatic (`ramsey_enc.py`, function `check_coloring`). This verifier intentionally shares no code with any search routine, and its logic is the definition itself; it was validated on the classical cases $R(3, 3) = 6$ and $R(4, 4) = 18$.

3.2 Exhaustive enumeration of cyclic colourings

For a cyclic colouring given by red classes A , an r -subset S is red-monochromatic iff the union of the chord classes of its edges is contained in A . We compute the set of chord-class masks realised by the r -subsets (and, separately, by the s -subsets) and perform a subset (zeta) transform over the 2^h masks; a colouring A is (r, s) -free iff neither A nor its complement contains any realised r -mask, respectively s -mask, as a subset. This determines the answer for the whole cyclic family in $O(2^h \cdot h)$ plus $O(\binom{n}{r})$ mask construction (`cyclic_ramsey.py`, SHA-256 in Table 4). This routine is fully general in (r, s) and is the method used for both families reported here.

3.3 Independent certification of the negative results

An enumeration that reports “none exists” is only as trustworthy as its implementation, so the same statements are re-derived by a method that shares no code with it. The cyclic problem is encoded as a SAT instance in the h Boolean variables “chord class d is red”, with the two clauses $\bigvee_{d \in C(S)} \neg x_d$ and $\bigvee_{d \in C(S)} x_d$ for each realised mask $C(S)$ of a (mono-free) r - or s -subset S (duplicate masks omitted). The instances are decided by CDCL and a DRAT proof is emitted, then checked by the independent proof checker `drat-trim` (`cyclic_sat.py`, general in (r, s, n) ; instances in Table 1 and 2; checker commit in Table 4).

3.4 One-vertex extension

Given a verified (r, s) -free colouring of K_n , adding a vertex w amounts to choosing its red neighbourhood Y , subject to: no red $(r - 1)$ -clique is entirely red towards w , and no blue $(s - 1)$ -clique is entirely blue towards w ; every monochromatic clique then lies in a copy already excluded inside K_n . This is a SAT instance with n variables. Unsatisfiability of this instance, together with a validated DRAT proof, is a machine-checkable certificate that the given graph is maximal among (r, s) -free graphs (`emit_cnf.py`, `certify_ext.py`).

3.5 Searches without certificates

Two incomplete or resource-bounded searches were also run; their outputs are reported in Section 7 as non-results: a CDCL portfolio with random vertex relabelling and solver diversification (32 workers, 12 h each on K_{43}), and a vectorised simulated-annealing local search with an incrementally maintained count of monochromatic cliques (up to 24 worker processes; cost increment verified against brute force for $0, \dots, 300$ iterations in a unit test suite). Neither produced a certificate, and neither is used as evidence anywhere in this note.

4 Results

4.1 The (5, 5) cyclic family

Theorem 3. *For $n \in \{42, 43, 44, 45, 46, 47, 48\}$ no cyclic (5, 5)-free colouring of K_n exists. For $n = 41$ exactly 20 of the $2^{20} = 1\,048\,576$ cyclic colourings are (5, 5)-free (ten up to interchange of colours); the red classes of one of them are $\{2, 3, 4, 6, 7, 9, 10, 11, 15, 16\}$.*

Proof. Exhaustive enumeration over all 2^h cyclic colourings by the subset transform of Section 3.2, independently re-derived and certified by the CDCL encoding and DRAT proofs of Section 3.3 (Table 1); for $n = 41$ the claimed colouring is verified by brute force over all $\binom{41}{5} = 749\,398$ five-subsets. $\square$

n	h	cyclic colourings 2^h	distinct K5 masks (per colour)	exhaustive time (s)	verdict	certificate DRAT
41	20	1 048 576	8 434	6.5	20 free	(SAT model)
42	21	2 097 152	9 061	7.7	none	verified
43	21	2 097 152	10 437	8.5	none	verified
44	22	4 194 304	11 231	10.7	none	verified
45	22	4 194 304	12 749	11.6	none	verified
46	23	8 388 608	13 640	15.0	none	verified
47	23	8 388 608	15 456	16.7	none	verified
48	24	16 777 216	16 488	22.6	none	verified

Table 1: Cyclic (5, 5)-free colourings of K_n . Times are single-core wall-clock times on the verification machine (Intel i5-class laptop, Python 3.12, NumPy); the CDCL re-derivation and `drat-trim` validation run in under a minute per n .

Corollary 4. *There is no vertex-transitive (5, 5)-free graph on 43 vertices, and none on 47 vertices.*

Proof. A vertex-transitive graph of prime order is circulant; apply Theorem 3 to $n = 43$ and $n = 47$. $\square$

Remark 5. The corollary is what makes the computation relevant to the open case: any counterexample to the conjecture $R(5, 5) = 43$ — i.e. any (5, 5)-free graph on 43 vertices — must have a trivial or non-transitive automorphism group, so the highly symmetric candidates that are cheapest to test are eliminated by proof. The same computation says nothing about composite orders 42 and 44–48, where transitivity does not force circulantcy: for $n = 42$ the classical construction itself is not cyclic, and vertex-transitive (5, 5)-free graphs of composite order are not excluded by Theorem 3.

4.2 The (4, 6) cyclic family

Theorem 6. *For $n \in \{34, 35, 36, 37, 38, 39, 40\}$ no cyclic (4, 6)-free colouring of K_n exists.*

Proof. Identical pipeline to Theorem 3: exhaustive subset transform, independently re-derived by the CDCL/DRAT encoding (Table 2); every stated `drat-trim` verdict in the table is VERIFIED. $\square$

Corollary 7. *There is no vertex-transitive (4, 6)-free graph on 37 vertices.*

Proof. 37 is prime, so a vertex-transitive graph on 37 vertices is circulant; apply Theorem 6 at $n = 37$. $\square$

n	h	distinct masks 2^h	red masks	blue masks	clauses	proof lines	certificate DRAT
34	17	2^{17}	712	10 339	11 051	248	VERIFIED
35	17	2^{17}	789	12 085	12 874	285	VERIFIED
36	18	2^{18}	844	15 950	16 794	152	VERIFIED
37	18	2^{18}	933	19 263	20 196	333	VERIFIED
38	19	2^{19}	1 014	24 223	25 237	303	VERIFIED
39	19	2^{19}	1 085	28 803	29 888	299	VERIFIED
40	20	2^{20}	1 197	33 637	34 834	276	VERIFIED

Table 2: Cyclic $(4, 6)$ -free colourings of K_n : no cyclic $(4, 6)$ -free colouring exists for $34 \leq n \leq 40$. “Distinct masks” = all chord-class masks realised by an r - or s -subset; red/blue masks are the counts used in the SAT encoding.

Remark 8. The published 35-vertex $(4, 6)$ -free graph of Exoo admits an automorphism of order 4 with a fixed point; hence it is *not* circulant and not vertex-transitive, and in particular it belongs to no cyclic family. Theorem 6 therefore does not reproduce that construction. The known lower bound $R(4, 6) \geq 36$ does not come from the symmetric family that we are able to rule out. Consequently any witness for $R(4, 6) \geq 37$ (i.e. a $(4, 6)$ -free K_{36}) must likewise be asymmetric.

4.3 Maximality of the Exoo graph

Theorem 9. *The $(5, 5)$ -free graph on 42 vertices published by Exoo (1989) — in our export, 435 red edges, verified to contain no monochromatic K_5 — cannot be extended to a $(5, 5)$ -free colouring of K_{43} . Equivalently, that graph is maximal among $(5, 5)$ -free graphs.*

Proof. The extension instance of Section 3.4 has 42 variables and 2318 clauses (1148 red and 1170 blue K_4 ’s), CNF file `ext42.cnf` (SHA-256 in Table 4). It is unsatisfiable; DRAT proofs produced by three independent CDCL solvers were validated by `drat-trim` (Table 3). $\square$

Remark 10. This does not exclude *other* $(5, 5)$ -free graphs on 43 vertices: if a witness for $R(5, 5) \geq 44$ exists, its 42-vertex induced subgraphs are necessarily different from the published one.

instance	solver	proof lines	drat-trim	note
ext42	Glucose (via <code>python-sat</code>)	92	VERIFIED	three solver names, identical bytes
ext42	Lingeling	139	VERIFIED	separate derivation
ext42	MapleChrono	116	VERIFIED	separate derivation
ext42	CaDiCaL 1.0.3 / 1.5.3 / 3.0.0	191 / 0 / 131	not verified	proof export of the API not a valid DRAT

Table 3: Unsatisfiability certificates and their external validation for the extension instance. The CaDiCaL rows are reported for completeness: the DRAT text returned by the binding is not a valid proof for the checker, so no claim rests on it.

4.4 Re-verification of the published witnesses

Proposition 11. *As part of this package the two published witnesses were re-verified from scratch with the independent brute-force checker of Section 3.1: Exoo’s 42-vertex graph is $(5, 5)$ -free (no 5-subset is monochromatic), and Exoo’s 35-vertex graph is $(4, 6)$ -free (no red K_4 and no blue K_6).*

Proof. `verify_exoo.py` loads the edge lists and runs `check_coloring` over all $\binom{42}{5} + \binom{35}{4} + \binom{35}{6}$ subsets; both flag `verified=True`. Edge lists and digests are in Table 4. These are *re-verifications of published data*, not new constructions; they anchor the verifier on objects from the literature. $\square$

5 Verification and reproducibility

This note ships a complete package of artefacts under `anc/` in the arXiv source, and the claims rest on them. The same set is mirrored in a versioned open repository, <https://github.com/kamil-libront/ramsey-certified-artefacts>, and — as a durable record intended to outlive the preprint — the CNF instances and DRAT certificates are also deposited in a permanent DOI repository (Zenodo); the links are inserted at submission. If arXiv’s upload-size limit is exceeded, the deposit is the canonical location and `anc/` holds only a pointer to it. Every artefact was produced locally and checked locally; no data left the machines. The full set, with SHA-256 digests and the exact commands, is in the companion `REPRODUCE.md` (English); here we record the principle and the headline commands.

Verification protocol. A reader who wants to confirm the claims needs no access to our machines, and in fact should *not* trust them. The protocol is: (1) recompute the enumeration of a cyclic family (Section 3.2, seconds); (2) re-derive the same statement from the shipped CNF with `drat-trim` (Section 3.3, seconds to a minute); (3) re-run the brute-force checks of the positive colourings (Section 3.1); (4) verify the DRAT proofs of Theorem 9. Steps 2 and 3 use checkers external to our search and enumeration code, so a hardware or implementation error in our layer cannot propagate into a claimed verdict.

artefact	SHA-256 (first 16 hex digits)
Exoo 42-v (5, 5)-free, edge list (435 red)	92549d13efc6dbfb
Exoo 35-v (4, 6)-free, edge list	ad43f25945f5eef7
<code>cyclic_ramsey.py</code> (zeta enumeration)	269a88b09eb6c23d
<code>cyclic_sat.py</code> (CDCL + DRAT encoding)	3c08163c75a31eb1
<code>emit_cnf.py</code>	23da360b...
<code>certify_ext.py</code>	73f56b36...
<code>extend_ramsey.py</code>	1b150c65...
<code>ramsey_enc.py</code> (independent brute-force verifier)	bba3da8e1023aeae
<code>ext42.cnf</code> (42 vars, 2318 clauses)	2b5735a4d1cd4056
<code>cyc_42.cnf</code>	5dd03b68...
<code>cyc_43.cnf</code>	af966485...
<code>cyc_48.cnf</code>	f13c0d8e...
<code>cyc_42.drat</code>	e7db6b9d...
<code>cyc_43.drat</code>	4e27cbd9...
<code>cyc_48.drat</code>	30691b16...
<code>cyc46_34..40.cnf</code> and <code>cyc46_34..40.drat</code> (VERIFIED)	see <code>REPRODUCE.md</code> 683c7cf5...
DRAT proofs of <code>ext42</code> (Glu/Ling/Maple)	2c6d2dd1... 41886e4c...

Table 4: SHA-256 digests of the inputs and certificates; the full digests, together with scripts and logs, are in the companion package. The checker `drat-trim` was built from commit `2e3b2dc0` (2024-11-25).

task	command
cyclic exhaustion	<code>python3 cyclic_ramsey.py 5 5 <n> (and 4 6 <n>)</code>
cyclic certification	<code>python3 cyclic_sat.py <r> <s> <n> glucose4 cyc_<n>.cnf cyc_<n>.drat</code>
proof check	<code>drat-trim <f>.cnf <f>.drat → VERIFIED</code>
extension instance	<code>python3 emit_cnf.py 5 5 exoo42.txt ext42.cnf</code>
extension certificate	<code>python3 certify_ext.py 5 5 exoo42.txt glucose4 prova.drat</code>
witness check	<code>python3 ramsey_enc.py check 41 red41.txt 5 5</code>

Table 5: Commands needed to reproduce and independently check every result in this note. Running the checks (right-hand side) is all a sceptical reader needs to do.

Hardware. Searches ran on a two-socket workstation ($2 \times$ Intel Xeon E5-2696 v3, 36 cores / 72 threads, 256 GB RAM, Windows 11); the enumeration, certification and proof checking reported here ran on a laptop (4 cores, Linux). **Software.** Python 3.11–3.12, NumPy, `python-sat` (Solvers: CaDiCaL, Glucose, Lingeling, MapleChrono, MapleSAT, MergeSAT), `drat-trim` (commit 2e3b2dc0). Any of the shipped proofs can be regenerated on any machine with `python-sat` and re-checked with `drat-trim`.

6 Prior work and positioning

The bounds $43 \leq R(5, 5) \leq 46$ and $36 \leq R(4, 6) \leq 40$, the Exoo constructions of 1989 and 2012, the conjecture $R(5, 5) = 43$, and the Angeltveit–McKay upper bound are standard; see Radziszowski’s dynamic survey and the references therein. Recent verified-computation methodology for Ramsey numbers — SAT + computer-algebra certificates in the style of Bright and co-workers — is the template we follow for the DRAT layer: the point is that an exhaustive search only becomes a mathematical statement once an independent checker confirms a machine-readable proof of the verdict.

Remark 12. The statements here are about *objects* (cyclic families and one specific graph) and are verified independently; whether the exhaustion of cyclic families and the maximality claim are already recorded in the literature is orthogonal to their correctness. Where they overlap known results, this note stands as an independent, certified reproduction; where they do not, they are new (small) computable facts. No claim in the note depends on novelty.

7 Limitations and non-results

We record explicitly what we did *not* establish, so that nothing here is over-read.

1. **No verdict on K_{43} .** A CDCL portfolio (32 workers with random vertex relabelling and five solver configurations) ran 12 h per worker on the $(5, 5)$ -free instance for $n = 43$ — approximately 384 CPU-hours — without deciding it. Reported as no-verdict, not as evidence for either side.
2. **Local search does not yet reproduce known constructions; we report this as a calibration failure, not a finding.** Our vectorised annealing search maintains an incrementally verified clique objective (mono-red K_r plus mono-blue K_s). On K_{35} for $(4, 6)$ that objective plateaus at best cost ≈ 816 (objective value $4 \cdot \#(\text{red } K_4) + 6 \cdot \#(\text{blue } K_6)$; a $(4, 6)$ -free colouring has value 0) and on K_{41} for $(5, 5)$ at ≈ 4350 ; the cost increment was verified against brute

force, so the plateau is not an arithmetic artefact. Twelve parameter variants (temperature, perturbation strength, kick, stagnation thresholds) all plateau at the same value — a *structural* plateau of the clique-only objective, not a tuning failure. Adding the induced- P_4 term that underlies Exoo’s method (weight λ , maximised) and sweeping $\lambda \in \{0, 8, 20, 40\}$ with 20 workers and 1800 s each also fails to reach cost 0 (best ≥ 1056). Because this search cannot recover objects known to exist, its “no solution” outcomes carry no weight; we use it only as a calibration instrument. This is the main methodological lesson of the exercise: *a search that has not passed positive controls cannot be quoted as evidence*.

3. **No DRAT certificate for CaDiCaL runs** through the `python-sat` API; the valid certificates in Table 3 come from other solvers. A clean solution would use CaDiCaL’s native proof output followed by `drat-trim` or an LRAT checker.
4. **Scope.** Theorems 3, 6 and 9 concern cyclic colourings and one specific graph. None is a counting argument settling $R(5, 5)$ or $R(4, 6)$; they prune candidate families and delimit the symmetric part of the search space.

What would be needed to settle $n = 43$. Either (i) a $(5, 5)$ -free colouring of K_{43} , which would refute the standard conjecture and which the search results above suggest is not reachable by naive annealing from the structures we tried; or (ii) an unsatisfiability certificate for the 903-variable instance, which at present requires cube-and-conquer with per-cube DRAT/LRAT proofs in the style of the $R(5, 5) \leq 46$ computation, i.e. cluster-scale resources. A single-machine contribution in that direction would be a certified unsat of a structured *restriction* (as here), not of the full instance.

8 Conclusion

This note delivers a package of structural facts about $(5, 5)$ - and $(4, 6)$ -free graphs, each backed by an independently checkable artefact: the cyclic families are empty in the stated ranges (so vertex-transitive witnesses on 43, 47 and 37 vertices do not exist), Exoo’s 42-vertex graph is maximal, and the two published witnesses are re-verified from scratch. The negative statements carry DRAT proofs validated by an external checker; the positive colourings are re-checked by brute force. The open cases remain open: we report no verdict on $R(5, 5)$ and $R(4, 6)$, the 384-CPU-hour portfolio returned nothing, and we report the calibration failure of our local search explicitly so that it is not mistaken for a result. A reader needs only the commands in Table 5 to confirm every claim independently.

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